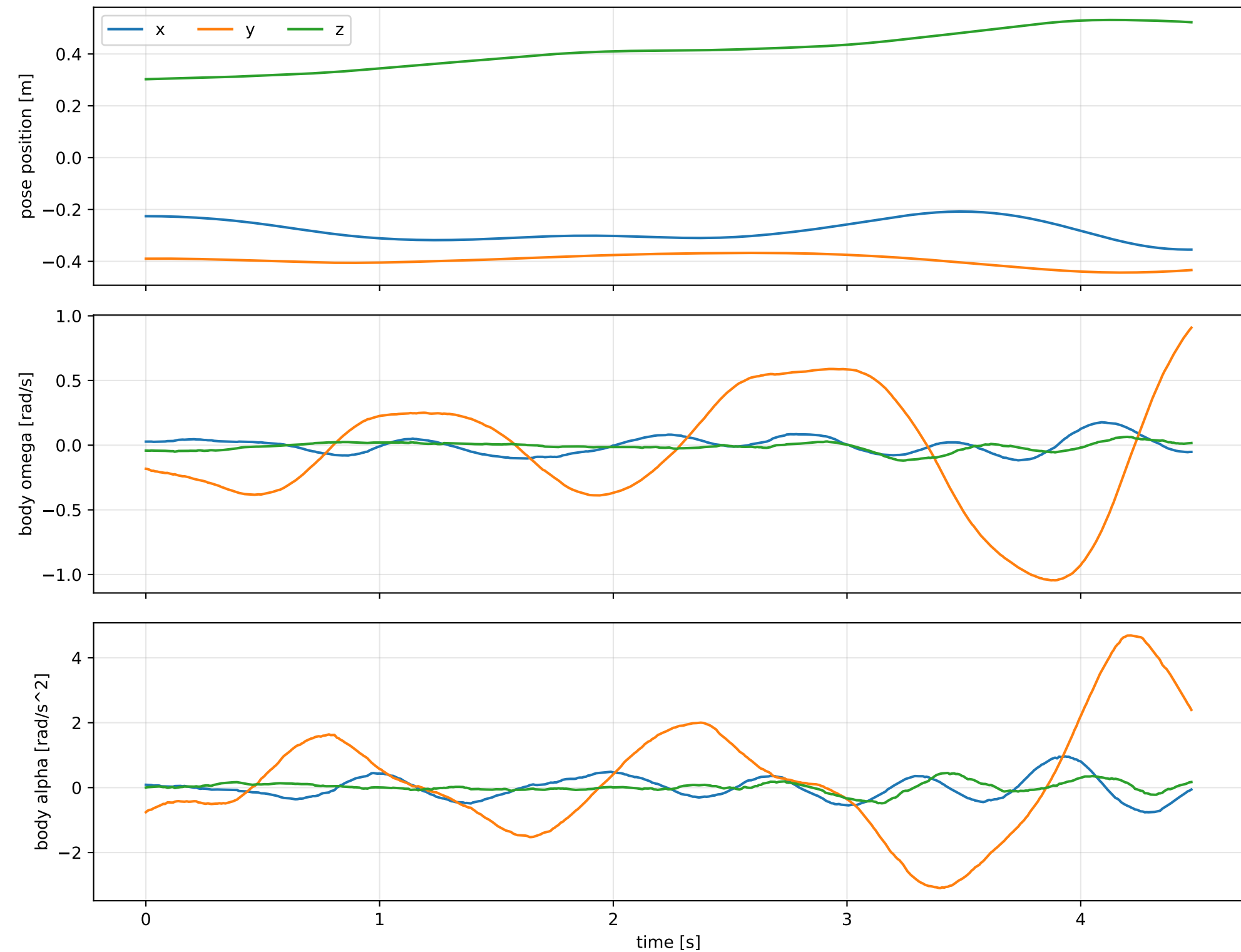
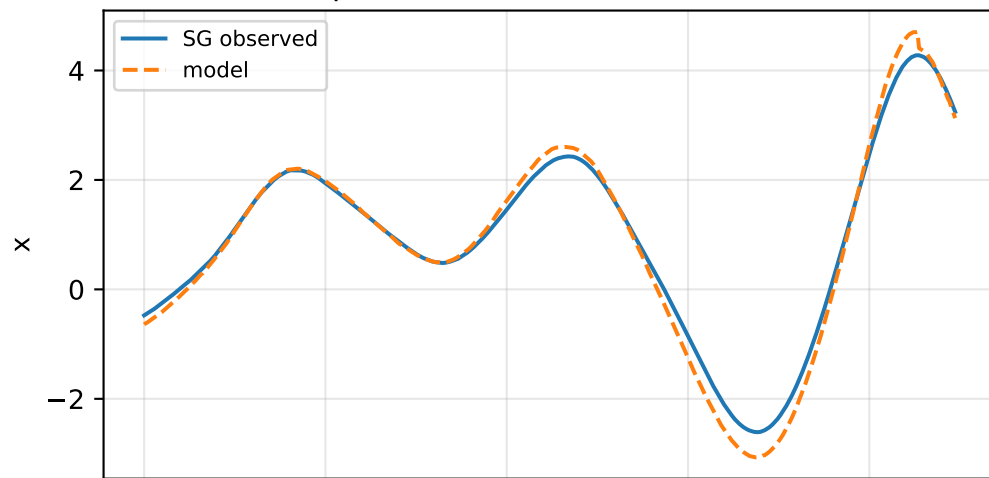


lag_fixed_one_period: SG trajectory and derived motion

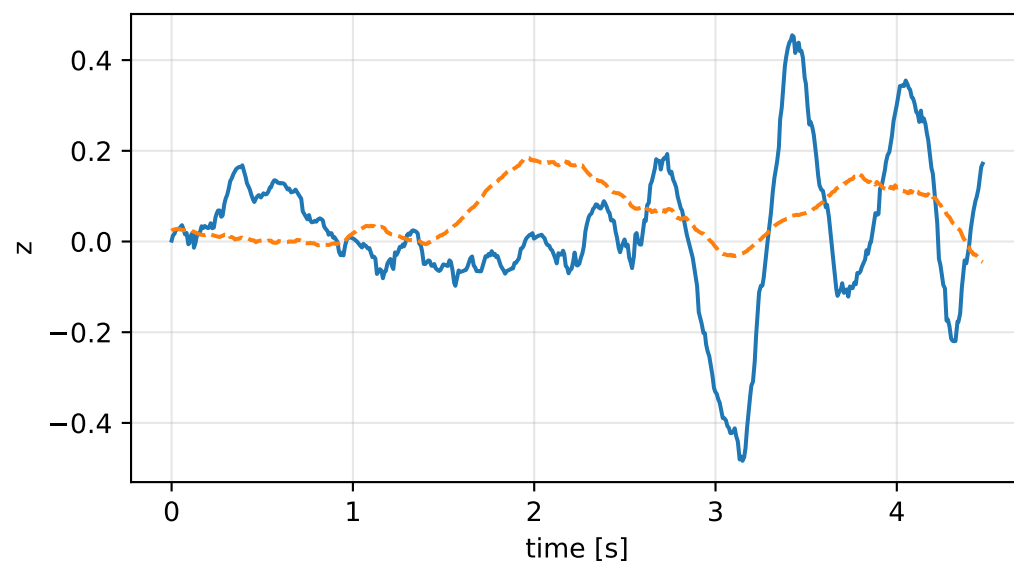
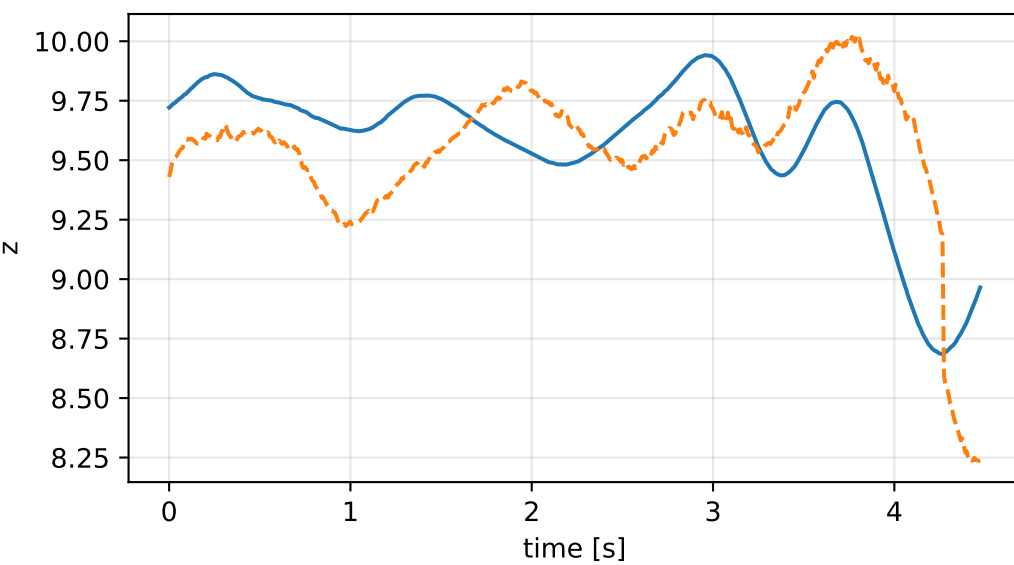
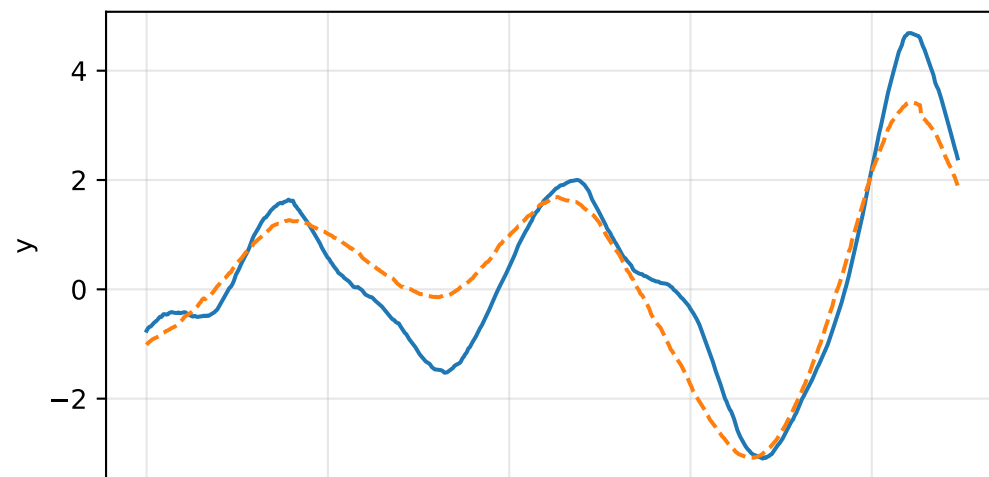
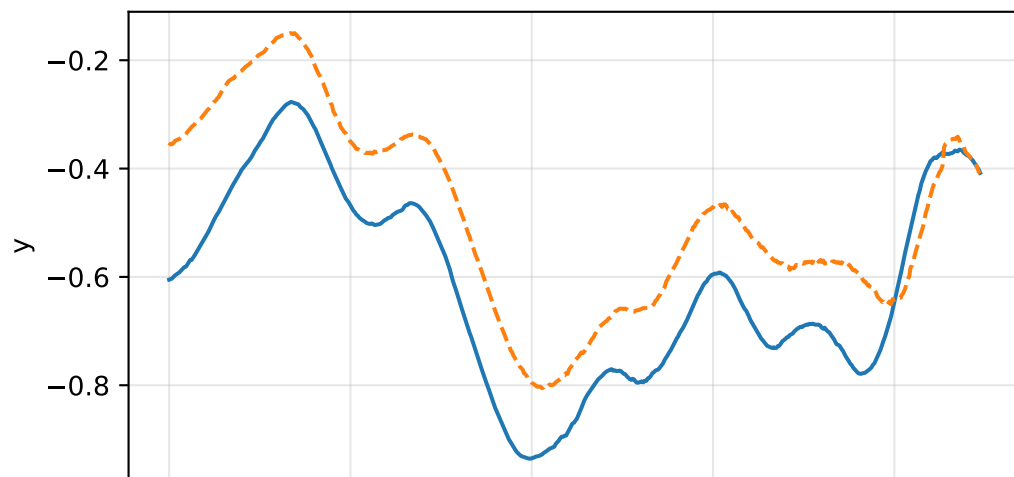
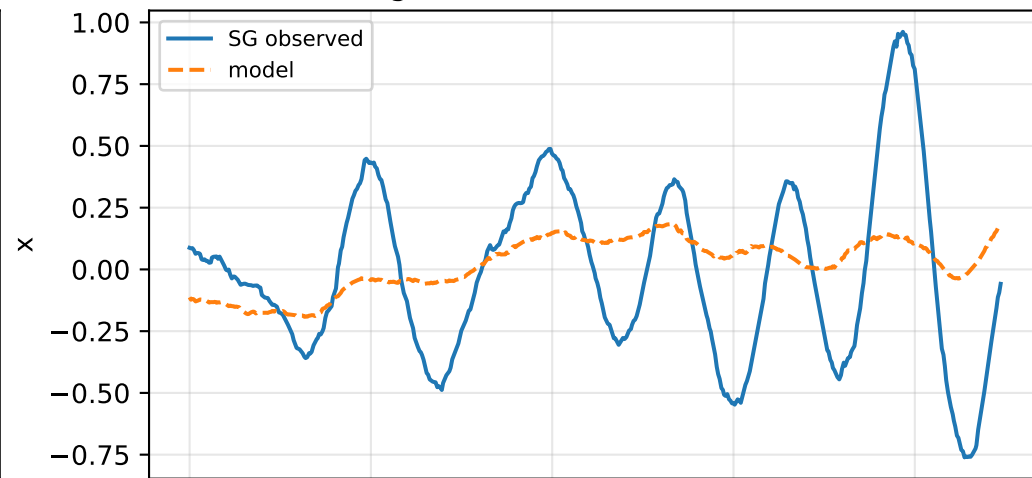


lag_fixed_one_period: acceleration residual objective

specific acceleration [m/s^2]

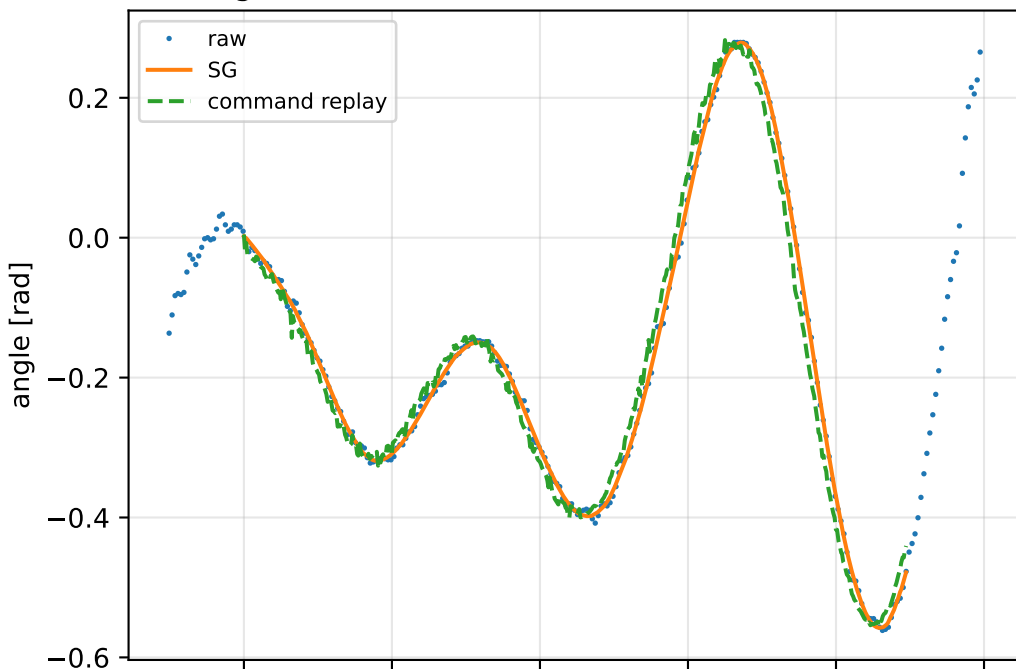


angular acceleration [rad/s^2]

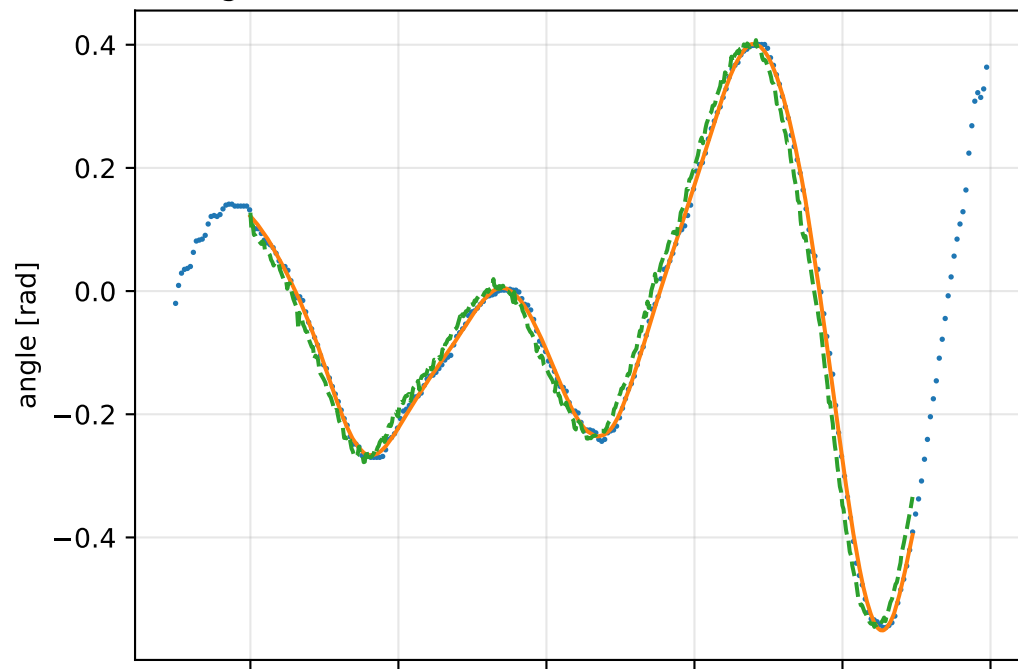


lag_fixed_one_period: gimbal measurement audit (objective=measured_sg)

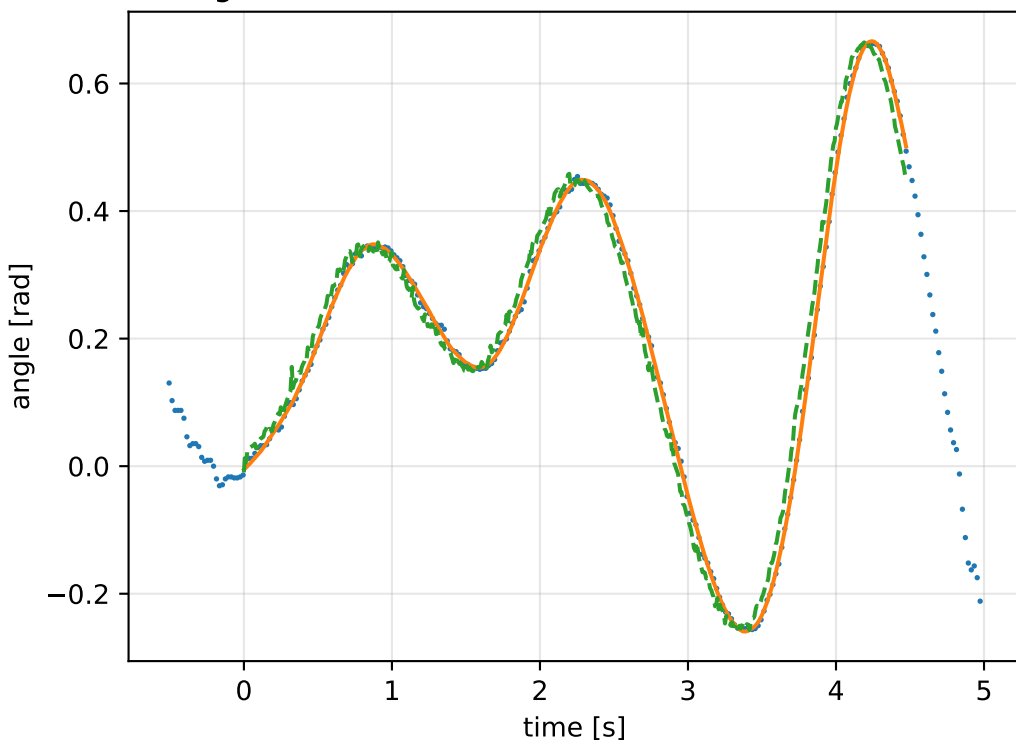
gimbal 1: RMSE=0.027, max=0.07529 rad



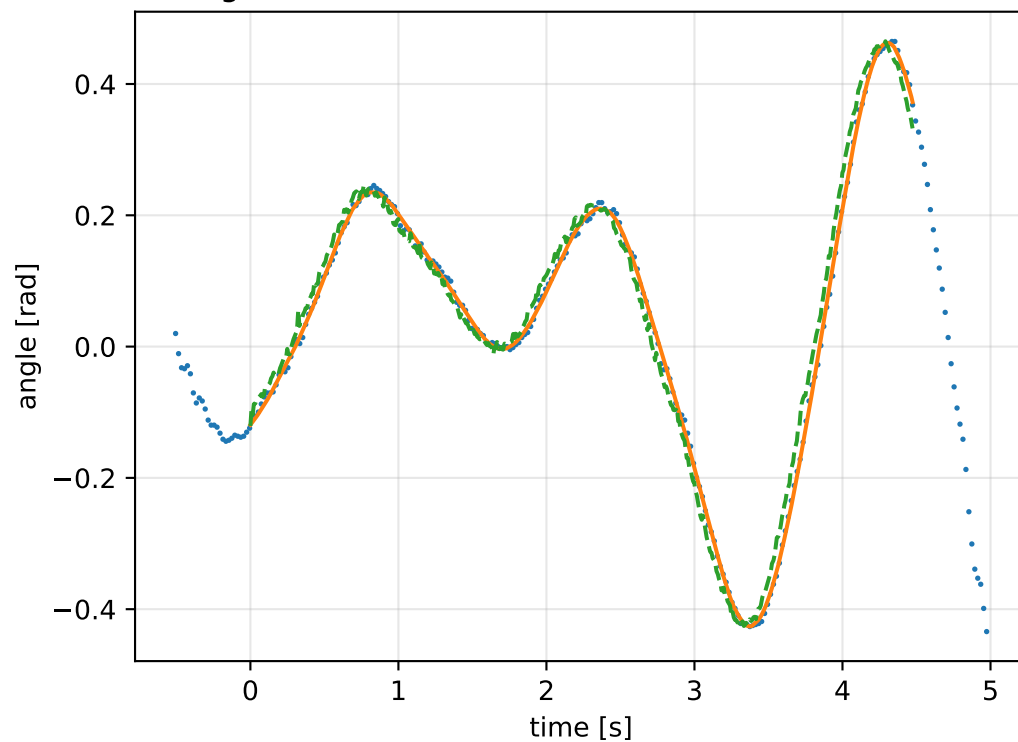
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

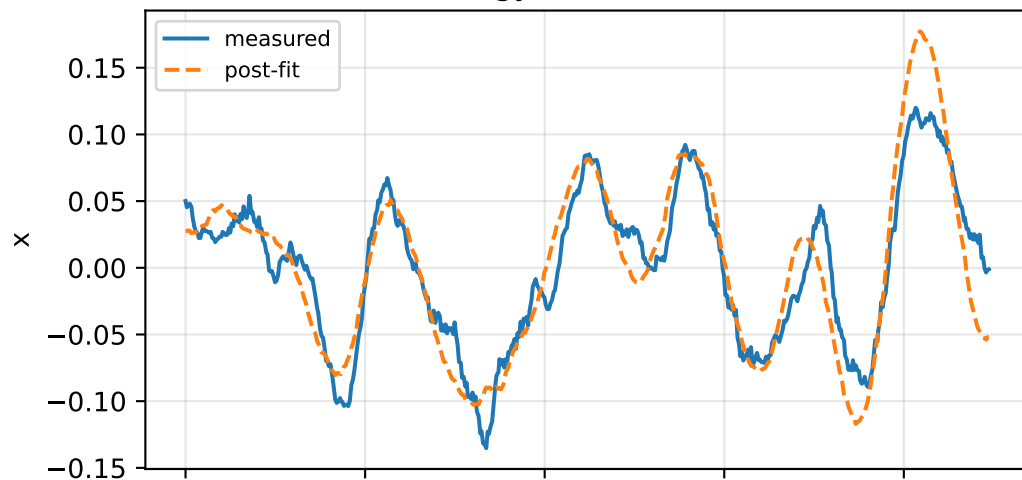
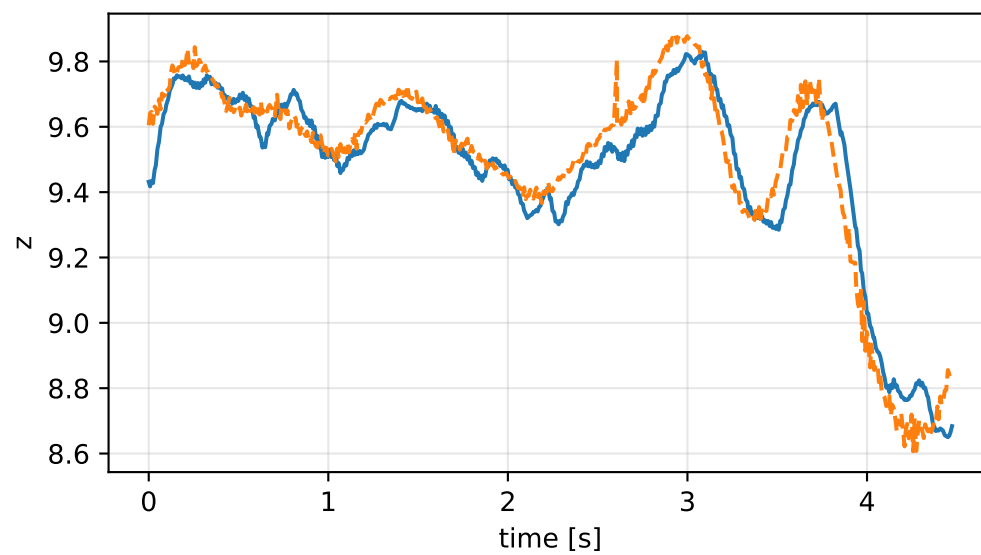
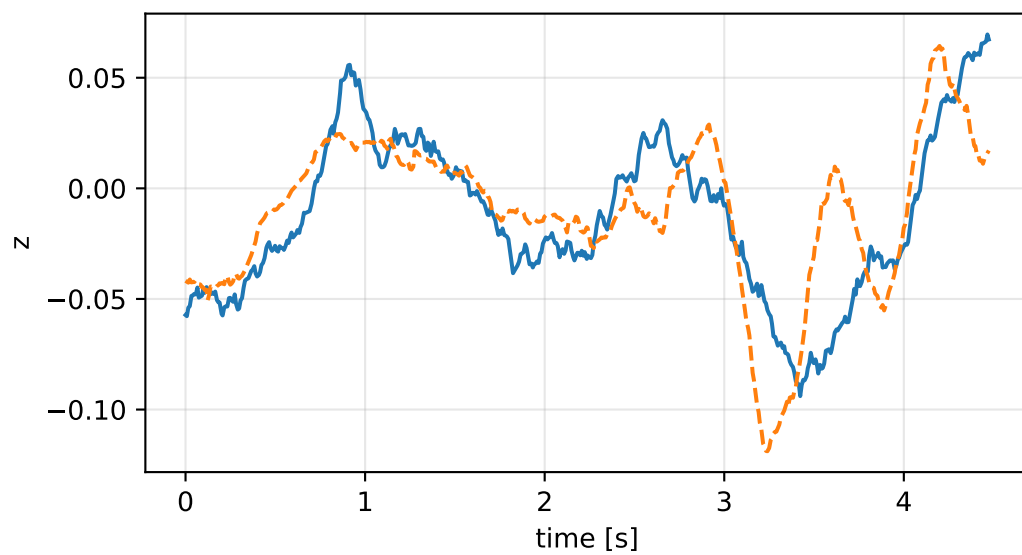
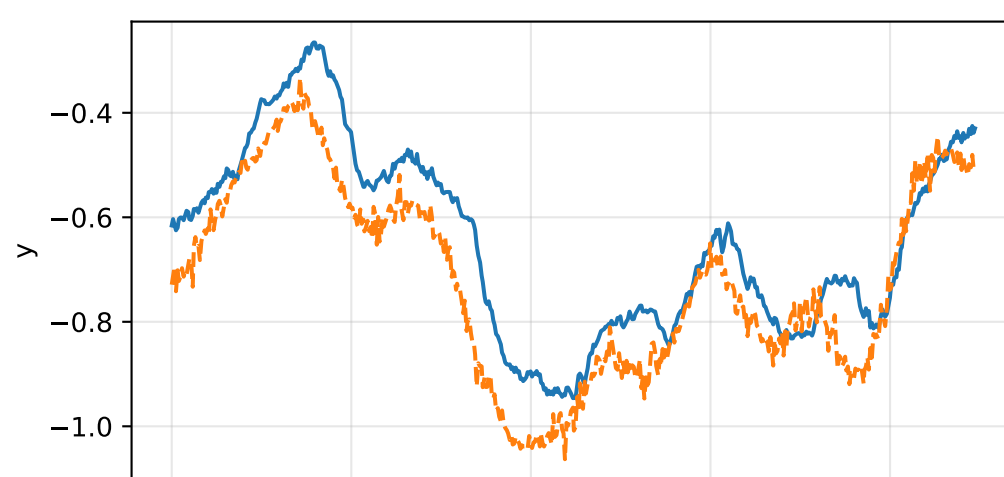
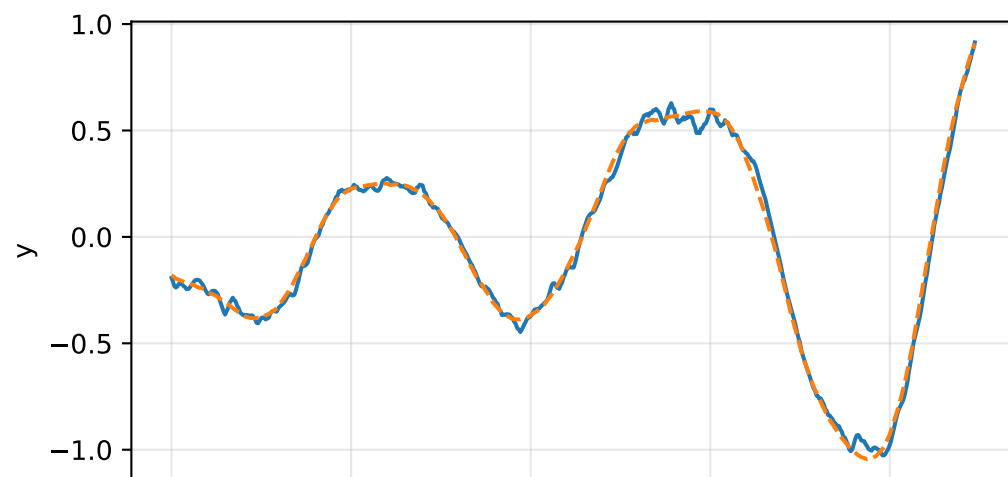
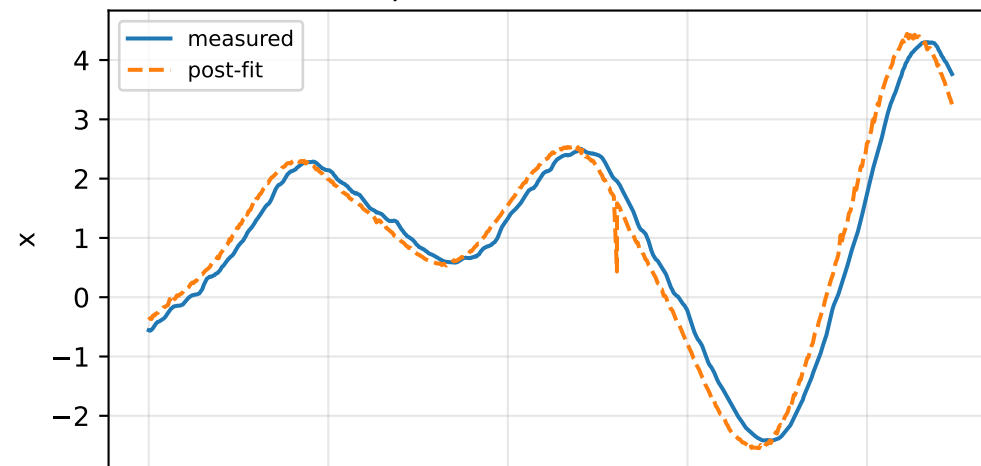


gimbal 4: RMSE=0.02598, max=0.07261 rad

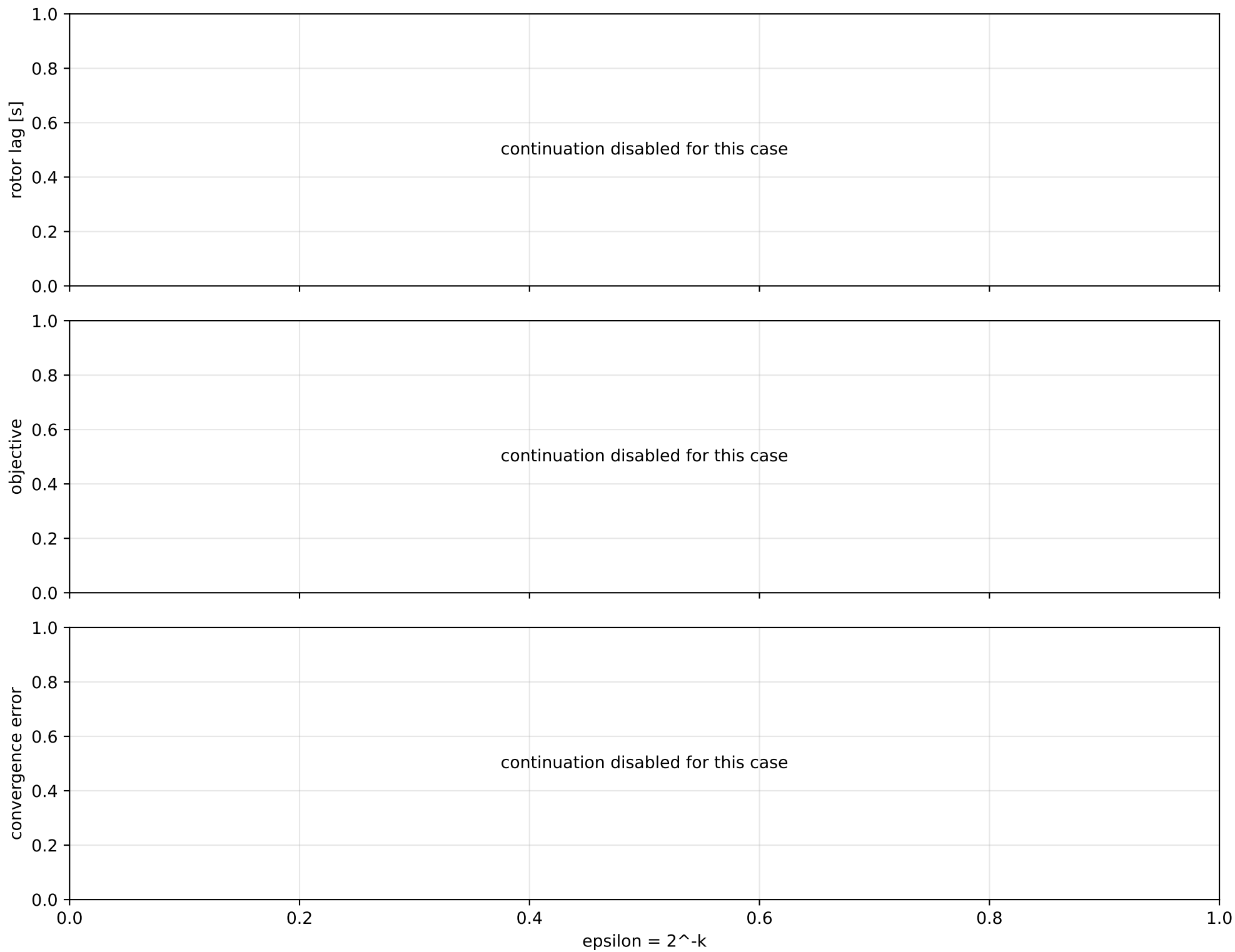


lag_fixed_one_period: IMU comparison (diagnostic only)

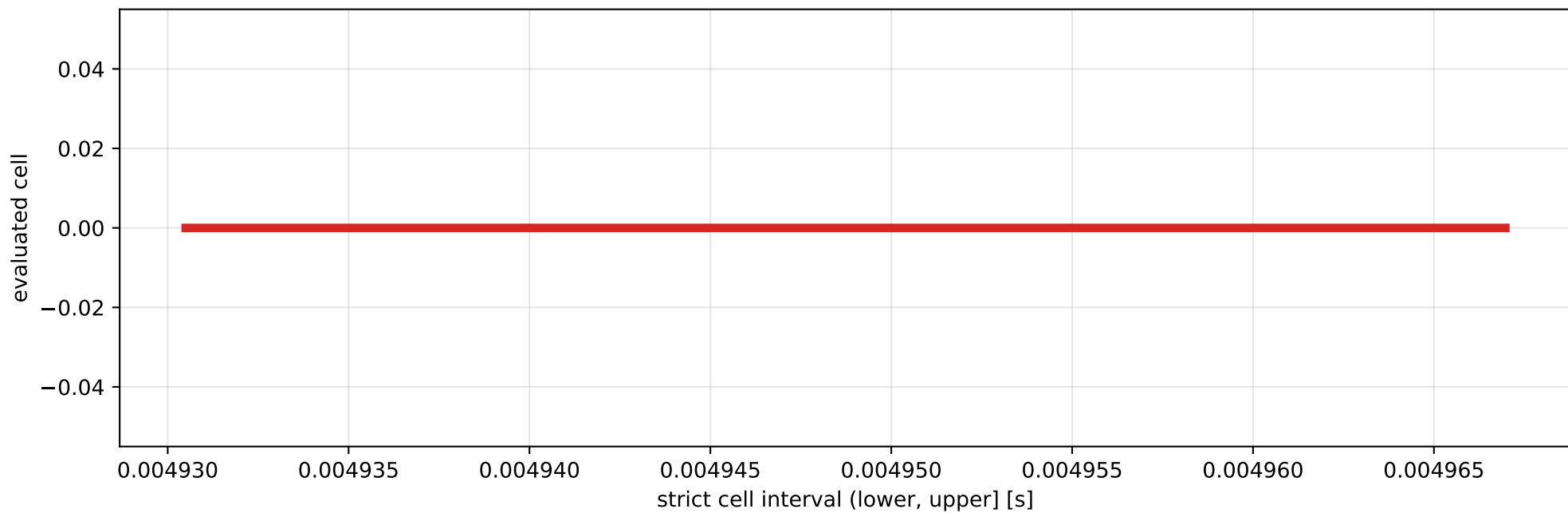
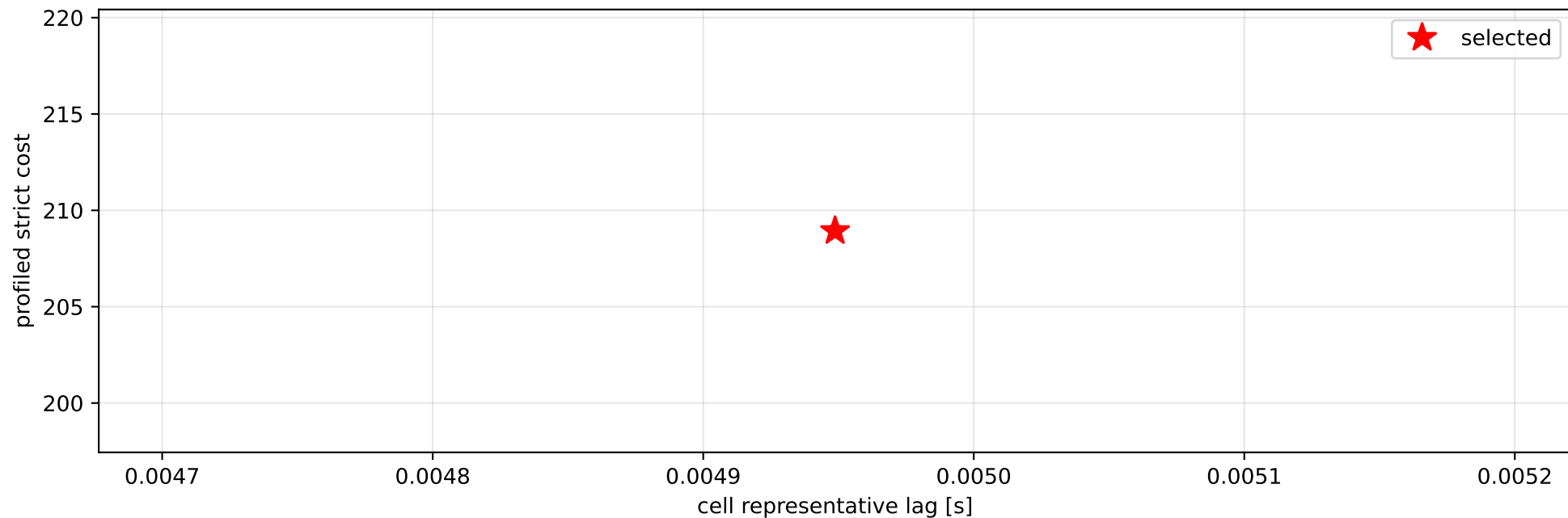
gyro [rad/s]

specific force [m/s²]

lag_fixed_one_period: smooth-to-strict continuation



lag_fixed_one_period: exact strict-ZOH lag cells



selected (0.00493049622, 0.00496697426] s; final smooth 0.00495767593 s; data boundary=False

lag_fixed_one_period: parameter estimate

Case: lag_fixed_one_period

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 11.7103064

inertia [kg m²]:

```
[[ 4.1290153 -0.15001615 -0.04031027]
 [-0.15001615  1.99480523 -0.7364205 ]
 [-0.04031027 -0.7364205  5.85197283]]
```

CoG body [m]: [-0.17008001 0.07868186 -0.1692834]

force effectiveness: [5.69065960e+00 1.93492928e-05 2.44984136e+00 8.29504508e+00]

scale-free J/m [m²]:

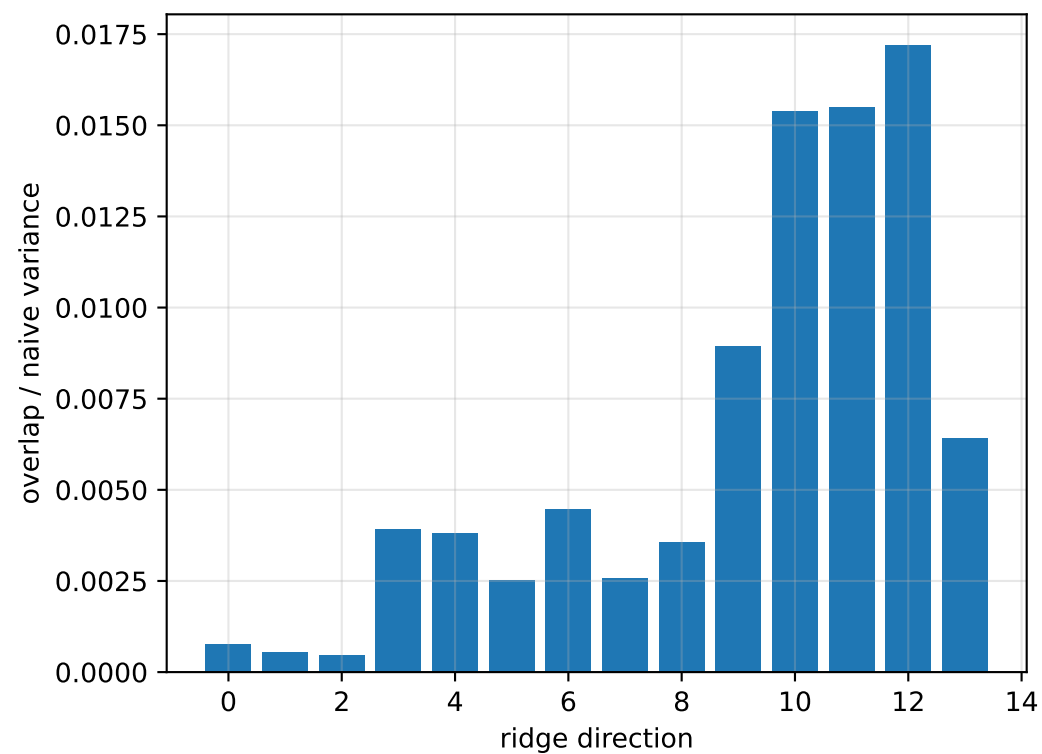
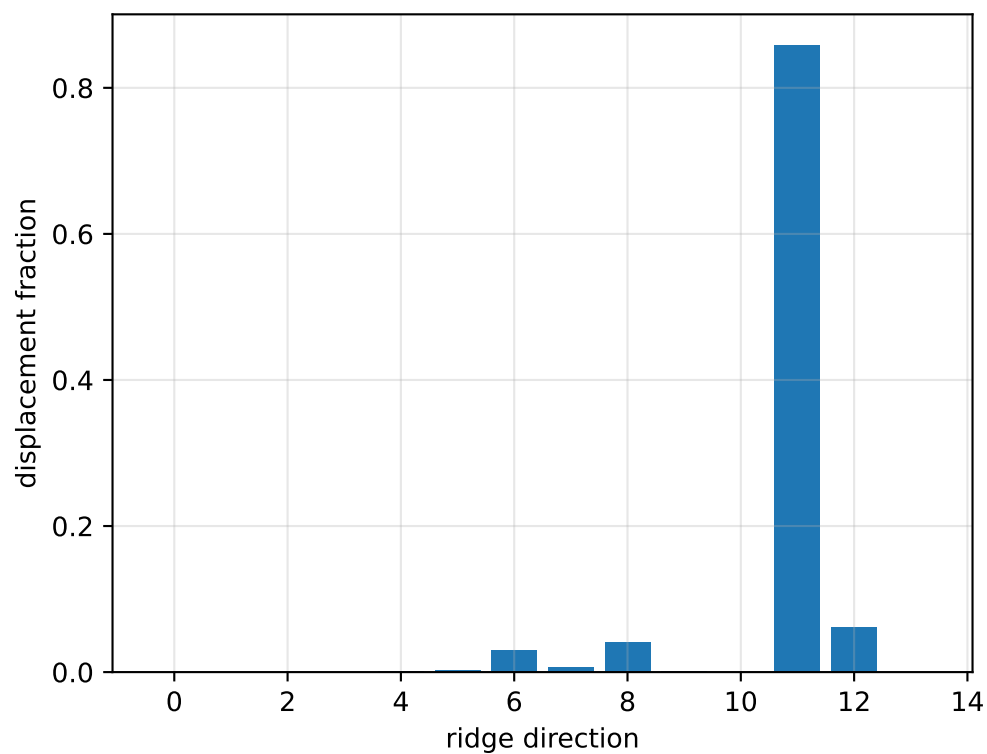
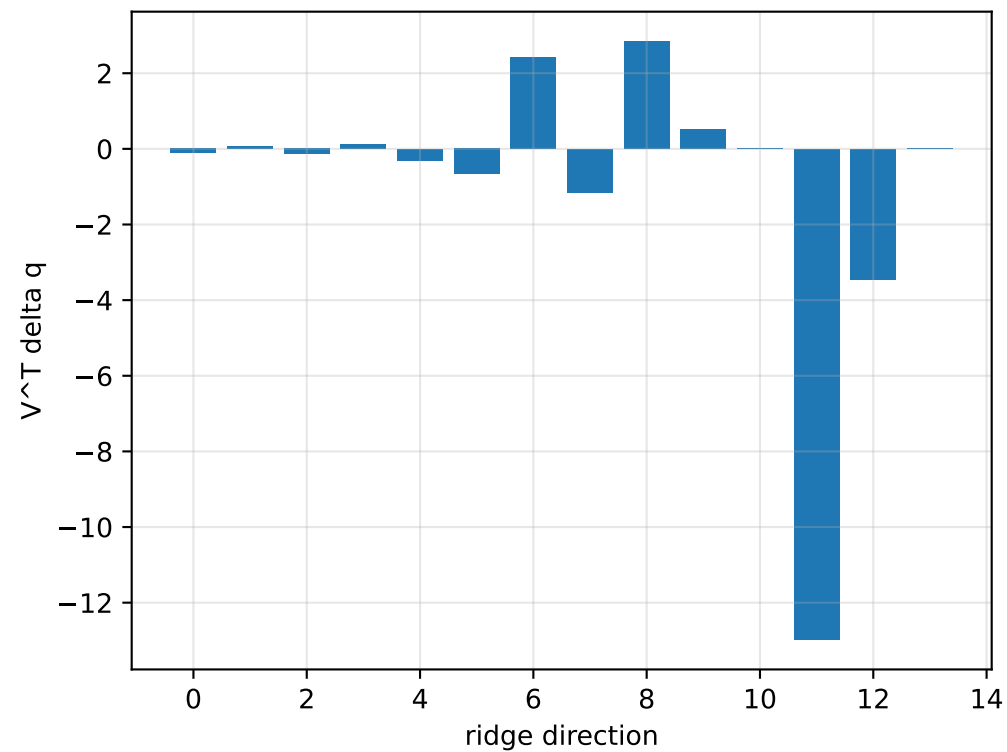
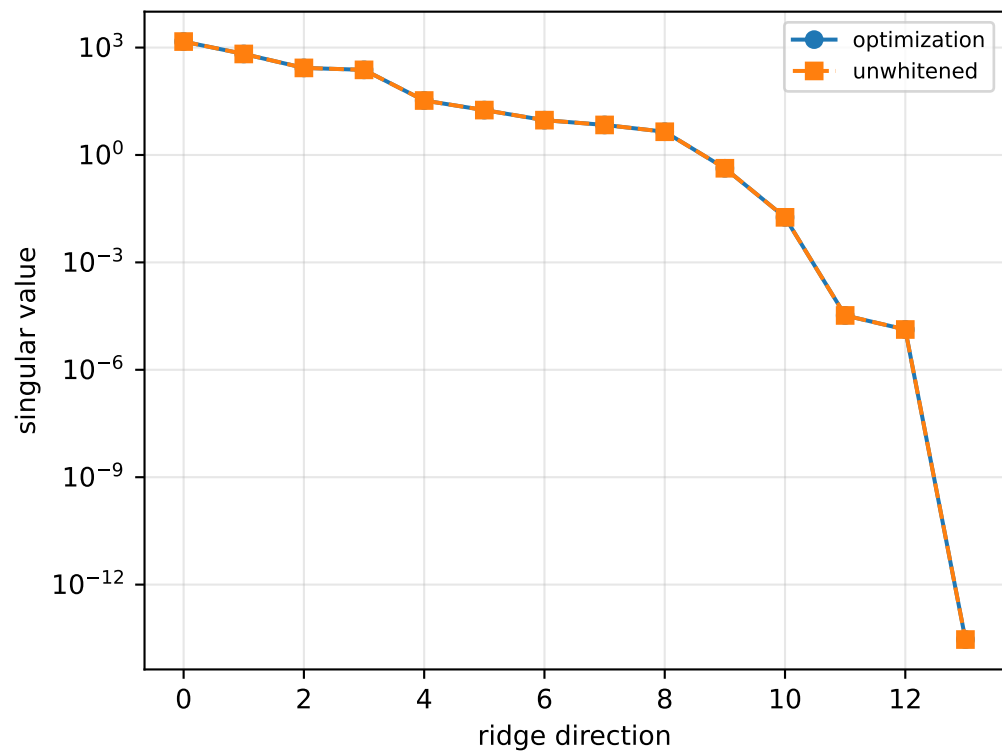
```
[[ 0.35259669 -0.01281061 -0.00344229]
 [-0.01281061  0.17034612 -0.06288653]
 [-0.00344229 -0.06288653  0.49972841]]
```

scale-free f/m: [4.85953090e-01 1.65233018e-06 2.09203864e-01 7.08354229e-01]

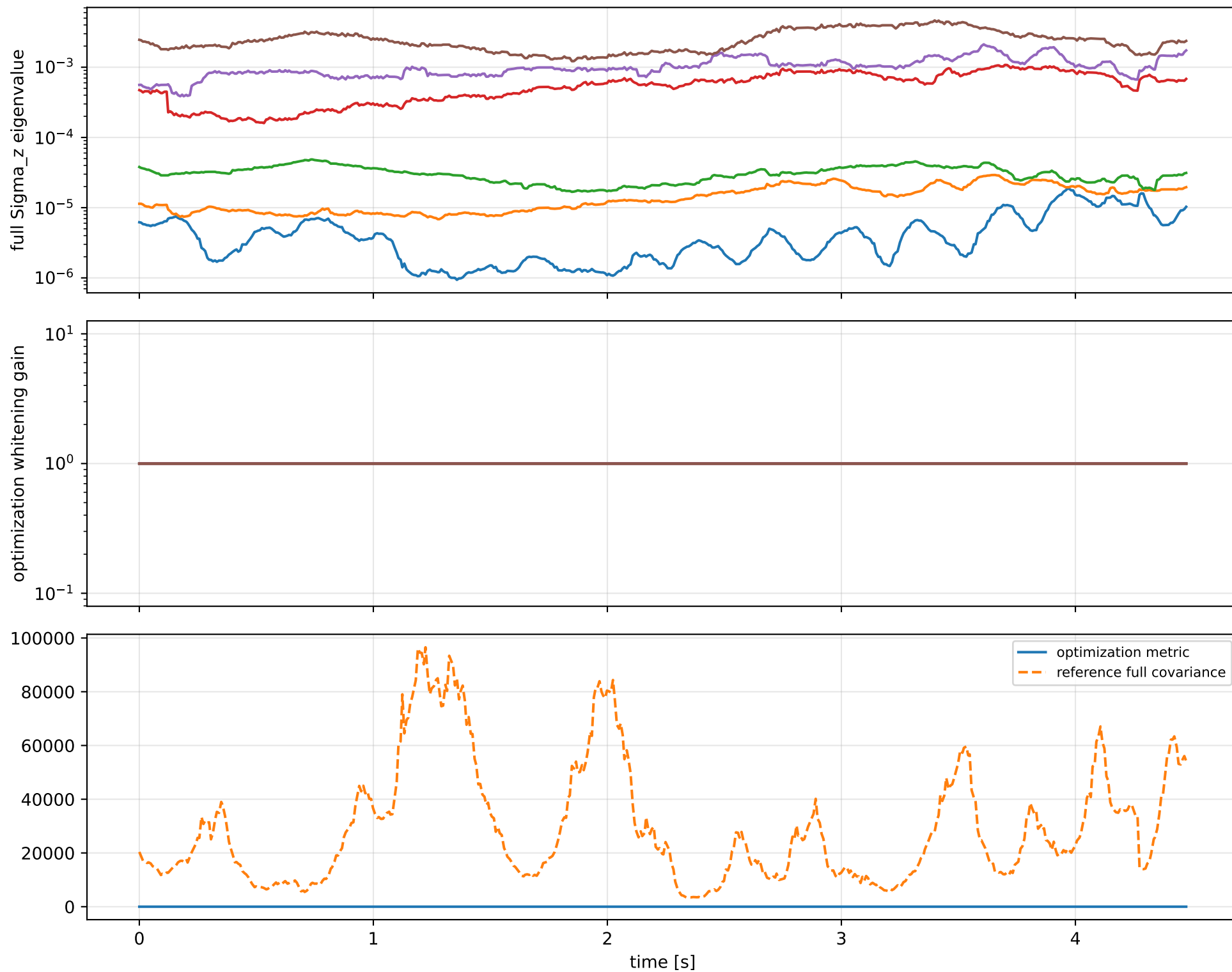
rotor lag cell: (0.00493049622, 0.00496697426] s

representative: 0.00495767593 s

lag_fixed_one_period: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=6.02\text{e-}14$)



lag_fixed_one_period: optimization vs reference covariance



lag_fixed_one_period: wrench / closure (force max $3.29\text{e-}14$, torque max $2.66\text{e-}15$)

